

Extended Quotient Block-Power Diagnostic

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Abstract

RH-245/RH-246 establish the exact orthogonal-quotient identity and a conditional block-power trace envelope, but the archived diagnostic stopped at dimension 512. We extend dense ordered-Schur calculations to dimension 1024, covering 23 endpoints and six new cases. Every quotient twelfth power is contractive. However the worst finite block root rate deteriorates from 0.393300 to 0.505642, the first contractive depth grows from seven to nine, and the unit-disk logarithmic tail diagnostic worsens by more than an order of magnitude. The mechanism survives finitely; uniform small-noise control remains open.

1 Exact quotient interface

Let E be the selected invariant generalized root space, let Π_E be its orthogonal projection, and put $Q = I - \Pi_E$. RH-245 proves that the compression $C = QAQ|_{E^\perp}$ realizes the quotient trace

$$\tau_n = \text{Tr}(C^n) \quad (1)$$

at every fixed finite operator [1]. The projection has norm one, although $E^\perp$ need not reduce A .

RH-246 supplies the exact block theorem: if for one integer m

$$\|C^m\| \leq \eta < 1, \quad \|C^m\|_1 \leq L_m, \quad \|C^r\| \leq K_r \quad (0 \leq r < m), \quad (2)$$

then all later traces have a geometric envelope with $q = \eta^{1/m}$ and an explicit logarithmic tail [2]. The theorem is exact; only the uniform hypotheses are missing.

2 Dimension-1024 protocol

For every archived endpoint of dimension at most 1024 we reconstruct the Hardy-scaled folded-Gaussian matrix, order its complex Schur form at the RH-222 radial cutoff, and split selected and quotient blocks. We then compute operator norms for powers one through twelve and the trace norm at power twelve. The extension contains 23 endpoints, six beyond the old cutoff.

The selected ranks match at all endpoints. The largest Schur trace-partition error is 3.99×10^{-15} ; the largest discrepancy with archived quotient traces is 4.76×10^{-9} at the new fine scales. These are floating diagnostics, not interval enclosures.

3 Finite deterioration result

quantity	RH-246 cutoff	present cutoff
maximum dimension	512	1024
endpoints	17	23
first contractive depth	3–7	3–9
contractive 12th powers	17/17	23/23
worst q_{12}	0.393300	0.505642
maximum $\ C^{12}\ $	1.37×10^{-5}	2.79×10^{-4}
unit-disk tail diagnostic	1.80×10^{-5}	5.65×10^{-4}

The worst new endpoint is the right channel at $\sigma = 0.0025$, dimension 1024. The finite geometric constant is $M_{12} = 551.5985$, while the rate worsens by a factor 1.28564 relative to RH-246.

Proposition 3.1 (Finite block contraction). *For each of the 23 audited quotient matrices, $\|C^{12}\| < 1$. Hence the RH-246 block theorem gives an all-order trace bound for each individual finite matrix using its audited constants.*

Proof. The maximum audited twelfth-power norm is $2.7932874477 \times 10^{-4}$. Substitute the finite matrix constants into (2). $\square$

Remark 3.2 (No uniform inference). Nine archived endpoints are still excluded by the dimension cutoff, and the worst constant deteriorates as the cutoff increases. The proposition is a family of 23 fixed-matrix statements, not a noise-uniform or continuum theorem.

4 Route decision

The signed cancellation retained by the quotient remains the correct mechanism, but finite extension gives no basis for extrapolating uniformity. RH-260 must combine this finite quotient tail with the exact analytic target tail and the still-obstructed anchored head. Gates A–E remain false/open. No Hilbert–Polya operator, Riemann-zero identification, zeta-divisor equality, or RH implication is claimed.

References

- [1] Bin Wang. Orthogonal-quotient superloop compression. *Preprint*, 2026. RH-245 preprint.
- [2] Bin Wang. Block-power quotient envelope criterion. *Preprint*, 2026. RH-246 preprint.